Human review packet: MBJG25ProducerFairness

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: MBJG25ProducerFairness
Paper id: MBJG25ProducerFairness
Source version: ICWSM 2025, 19(1), 1139-1157
Source record: audit/paper_statement_map.json
Rows: 2
Generated: 2026-09-06
Source URL: official source

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper MBJG25ProducerFairness --serve
```

Open http://127.0.0.1:8765/ in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

Paper-specific semantic prerequisites

MBJG25ProducerFairness.paper_fixed_binary_model_formulaSpec

Paper-local declaration:

MBJG25ProducerFairness.paper_fixed_binary_model_formulaSpec
Source locator: cited publication, lines 159-219

Verbatim paper-source connection

with binary reviews, our model extends readily to ordinal reviews as well (see Appendix E).

2.1 Model Primitives

Our modeling framework is as follows.

Products. There is a universe of products V. Product $v \in V$ has unobserved true quality q_v, where $0 \le q_v \le 1$.

Binary ratings that accumulate over time. Products accumulate ratings over time, as users interact with and rate products. To capture this phenomenon, we assume the market operates over discrete time periods t, and products can receive ratings at each time period. We assume these ratings are binary, i.e., 0 or 1.

Crucially, the true quality of a product should impact the rating received; we suppose that a rating for product v is Bernoulli(q_v), independent of all other randomness in the system. Since ratings accumulate, we use $S(v, T)$ to denote the set of binary ratings that product $v \in V$ has received after T timesteps have elapsed.

We assume binary ratings as a simplification for ease of exposition; our results qualitatively extend to other settings, as we demonstrate in Appendix E. We note that this choice captures how many platforms elicit ratings (e.g., in any platform that uses a thumbs-up/thumbs-down system); in other platforms with more than two options, ratings often follow a “J-curve” where users mostly only use the extreme options (Hu, Zhang, and Pavlou 2009). In Appendix G, we demonstrate that our results hold even when we assume users’ distribution over ratings is centered or even right-skewed.

Prior-Weighted Rating System. In a prior-weighted rating system, the prior distribution is a design choice of the platform. The system operates by estimating a posterior distribution of the estimated true quality for each product, given the prior and the received ratings on that product.

We primarily consider rating systems where the prior represents a Beta distribution.[1] Then, in this setting, the platform chooses the prior Beta distribution parameters $(\hat\alpha, \hat\beta)$.[2] We will refer to $\hat\alpha + \hat\beta$ as the prior strength, as it indicates how strongly the prior is concentrated around its mean (and how many ratings will be needed to “overwhelm”) the prior.

After a product receives a rating, the platform updates its estimation posterior distribution for its true quality. In our

1 This choice is natural, given that ratings are assumed to be Bernoulli. The Beta distribution is the conjugate prior for Bernoulli data, i.e., with this choice of prior, the posterior distribution remains a Beta distribution but with different parameters.

2 In a Beta(α, β) distribution, the mean is $\frac{\alpha}{\alpha+\beta}$; the higher that $\alpha + \beta$ are, the more concentrated the distribution around the mean. A common interpretation of the parameters is that $\alpha+\beta$ is the number of trials (coin flips), and α represents the number of successes.

is the number of negative ratings). We can compute the posterior mean estimated quality for product v at timestep t as:

$$\hat{q}_{\hat\alpha,\hat\beta}(v, t) = \frac{\hat\alpha + R(v, t)}{\hat\alpha + \hat\beta + |S(v, t)|}. \quad (1)$$

The platform impacts the operation of a prior-weighted rating system only through the prior choice, determined here by $\hat\alpha$ and $\hat\beta$. There are two aspects of this choice: the shape of the prior, and its strength relative to observed ratings in determining the posterior. In our analysis, we fix the prior shape by relying on market-level data, in a manner inspired by empirical Bayesian statistical methods (Robbins 1964). Our analysis shows that the strength of the prior then mediates the tradeoff between efficiency and producer fairness.

Formally, to reason about a prior’s strength, we fix a prior shape $(\tilde\alpha, \tilde\beta)$, then set $(\hat\alpha, \hat\beta) = (\eta\,\tilde\alpha, \eta\,\tilde\beta)$ for some prior strength $\eta \ge 0$, providing us with a single-parameter lever to adjust the strength of the prior. We elaborate on how we calibrate the prior shape in Section 3.2. Note in particular that if $\eta = 0$, then the platform simply implements sample averaging to compute the estimated quality of an item.

2.2 Fixed and Responsive Models

We study two models for how customers may interact with items, and items accumulate ratings over time. We begin with a fixed model in which the products receive ratings at the same rate, regardless of their past ratings. We then study a (more realistic) responsive model in which products with higher estimated ratings receive ratings more often.

Fixed model. In the fixed setting, all products stay in the market for the entire time horizon, and each product $v \in V$ accumulates one rating at each timestep t. In this model, the platform eventually accurately learns the true quality of all products, but has noisy estimates for each finite time t.

Responsive model. The responsive model differs from the fixed model in two ways: (1) items enter and exit the marketplace over time; (2) items are more likely to receive ratings if they have a higher estimated quality in the rating system. These changes in the responsive model capture selection effects (Acemoglu et al. 2022), in which consumers may react to previous ratings, more often choosing to interact with (and thus rate) items with more positive ratings in the past; these effects are not present in the fixed model.

At each timestep t, only a subset $M(t) \subseteq V$ of the products is available for purchase. A single consumer chooses a single product $v(t)$ to purchase and provides a rating on that product, and at the end of the timestep, each product $v \in M(t)$ exits the marketplace with some uniform, exogenous probability and is replaced by another product not currently in the market. The customer choice will depend

[Next verbatim source excerpt]

compromising anonymity...

(a) If your work uses existing assets, did you cite the creators? Yes; we cite the datasets used (Gao et al. 2022; Wan et al. 2019; Wan and McAuley 2018; Hou et al. 2024).

(b) Did you mention the license of the assets? Yes, when we first perform in-depth analysis for each dataset.

(c) Did you include any new assets in the supplemental material or as a URL? No, because we did not develop any new datasets or assets outside of our simulation experiments.

(d) Did you discuss whether and how consent was ob-

$$\text{Bias}^2 = (E[\hat{q}_{\eta\tilde\alpha,\eta\tilde\beta}(v, t)|q_v] - q_v)^2 \quad (6)$$

$$= \left(E\left[\frac{\eta\,\tilde\alpha + R(v, t)}{\eta\,\tilde\alpha + \eta\,\tilde\beta + t}\right] - q_v\right)^2 \quad (7)$$

$$= \left(\frac{\eta\,\tilde\alpha - \eta\,\tilde\alpha q_v - \eta\,\tilde\beta q_v}{\eta\,\tilde\alpha + \eta\,\tilde\beta + t}\right)^2. \quad (8)$$

Variance can also be broken down accordingly:

$$\text{Variance} = \text{Var}[\hat{q}_{\eta\tilde\alpha,\eta\tilde\beta}(v, t)|q_v] \quad (9)$$

" [U+0010 control character] [U+0011 control character]2 #

tained from people whose data you're using/curating? No, because the license for the dataset permits sharing and adaptation of the content so long as it is properly attributed.

(e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? No, because the original authors of the dataset fully anonymized it.

(f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR? N/A

(g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset? N/A

6. Additionally, if you used crowdsourcing or conducted research with human subjects, without compromising anonymity...

$$= E\left[\hat{q}^{\eta\tilde{\alpha},\eta\tilde{\beta}}(v,t) - E[\hat{q}^{\eta\tilde{\alpha},\eta\tilde{\beta}}(v,t)] \mid q_v\right] \quad (10)$$

$$E[(R(v,t) - E[R(v,t)])^2 \mid q_v]$$

$$= \frac{E[(R(v,t) - E[R(v,t)])^2 \mid q_v]}{(\eta\tilde{\alpha} + \eta\tilde{\beta} + t)^2} \quad (11)$$

$$= \frac{\mathrm{Var}[R(v,t) \mid q_v]}{(\eta\tilde{\alpha} + \eta\tilde{\beta} + t)^2} \quad (12)$$

$$= \frac{tq_v(1-q_v)}{(\eta\tilde{\alpha} + \eta\tilde{\beta} + t)^2}. \quad (13)$$

The expected MSE can be written as follows:

$$\mathrm{MSE} = \left(\frac{\eta\tilde{\alpha} - \eta\tilde{\alpha}q_v - \eta\tilde{\beta}q_v}{\eta\tilde{\alpha} + \eta\tilde{\beta} + t}\right)^2 + \frac{tq_v(1-q_v)}{(\eta\tilde{\alpha} + \eta\tilde{\beta} + t)^2}.$$

Lean-elaborated paper signature

type:
{alpha beta eta q : ℝ} → {t : ℕ} → 0 < alpha → 0 < beta → 0 ≤ eta → 0 < t → 0 ≤ q → q ≤ 1 → Prop

value:

```
fun {alpha beta eta q : ℝ} {t : ℕ} (halpha : 0 < alpha) (hbeta : 0 < beta) (heta : 0 ≤ eta) (ht : 0 < t) (hq0 : 0 ≤ q)
    (hq1 : q ≤ 1) =>
  MBJG25ProducerFairness.paper_expected_posterior_mean alpha beta eta t q =
     (eta * alpha + ↑t * q) / (eta * alpha + eta * beta + ↑t) ∧
    MBJG25ProducerFairness.paper_conditional_bias alpha beta eta t q =
       MBJG25ProducerFairness.paper_expected_posterior_mean alpha beta eta t q - q ∧
     MBJG25ProducerFairness.paper_conditional_variance alpha beta eta t q =
         ↑t * q * (1 - q) / (eta * alpha + eta * beta + ↑t) ^ 2 ∧
       MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta eta t q =
         MBJG25ProducerFairness.paper_conditional_bias alpha beta eta t q ^ 2
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The source specifies binary Bernoulli ratings in discrete timesteps, one rating per product per fixed-model timestep, Beta parameters (eta*alpha, eta*beta) with eta >= 0, and its Appendix-A bias and variance formulas. The target faithfully uses a positive natural review count cast to ℝ and states the corresponding conditional expectation, bias, variance, and squared-bias identities.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 313-334

Verbatim source input

```
Theorem 3.1. Suppose we have a prior-weighted rating sys-
tem in the fixed setting with prior parameters (η α̃, η β̃). Fix
product v with true quality qv and consider quality estima-
tion after t timesteps.
   Then, as η            ≥ 0 increases, (1) squared bias
[U+0010 control character]                [U+0011 control character]2
  E[q̂ηα̃,ηβ̃ (v, t)|qv ] − qv     is nondecreasing; (2) vari-
ance Var[q̂ηα̃,ηβ̃ (v, t)|qv ] is strictly decreasing in η.
   Theorem 3.1 establishes the trade-off between fairness
and efficiency. As the prior strength of a prior-weighted rat-
ing system increases, it is less likely for two identical prod-
ucts to be rated differently, reducing variance while increas-
ing estimation bias.
   The above theorem characterizes error for a given quality
level qv as prior strength changes. We are also interested in
how the true quality of the product influences the efficiency
and fairness metric, i.e., how the platform estimates quality
for different true quality levels. An ideal statistical estimator
for quality should have low MSE across all quality levels.
Our next theorem characterizes how efficiency and unfair-
ness vary with true product quality:

iments, is attached in the submission.
   Model calibration. To pick a prior shape (α̃, β̃), we make
use of the empirical Bayes (EB) method (Robbins 1964),
which provides an approach for computing an estimate of
prior parameters across a population. We perform a 60-40
train-test split on the products in the our dataset.
↪ We use all
available ratings data for products in the training set to cali-
brate (α̃, β̃) using EB. The remaining 1,331 products (in the
test set) are the universe of products V on which the simu-
lation is run. We are interested in investigating how ratings
in the marketplace change over time, so our marketplace is
simulated over time; we fix a finite time horizon T , and at
each timestep t, either draw one review for all v ∈ V in the
fixed model, or draw one review for the specific v ∈ V cho-
sen by the consumer at timestep t in the responsive model.
   Fixed model parameters. We set a time horizon of T =
50, and run our simulations for four different values of
η ∈ {0, 1, 10, 1000}. The first two η values correspond to
prior-weighted rating systems under the sample mean esti-
mator and the baseline empirical Bayes (EB) estimator re-
spectively. η = 10 corresponds to a setting where the prior
is difficult to overcome in early timesteps, but eventually is
```

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
Corrected Theorem 3.1: in the fixed Beta-Bernoulli model, after a positive discrete review count and for nonnegative prior strengths with etaLow <=
↪ etaHigh, squared conditional bias is nondecreasing; conditional variance weakly decreases for every true quality q in [0,1], and strictly decreases
↪ when 0 < q < 1 and etaLow < etaHigh.
```

Archival source anchor: cited publication, lines 313-320

Expanded PaperInterface specification

```
∀ {alpha beta q etaLow etaHigh : ℝ} {t : ℕ},
  0 < alpha →
    0 < beta →
      0 < t →
        0 ≤ q →
          q ≤ 1 →
            0 ≤ etaLow →
              etaLow ≤ etaHigh →
                MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta etaLow t q ≤
                    MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta etaHigh t q ∧
                  MBJG25ProducerFairness.paper_conditional_variance alpha beta etaHigh t q ≤
                      MBJG25ProducerFairness.paper_conditional_variance alpha beta etaLow t q ∧
                    (0 < q →
                      q < 1 →
                        etaLow < etaHigh →
                          MBJG25ProducerFairness.paper_conditional_variance alpha beta etaHigh t q <
                            MBJG25ProducerFairness.paper_conditional_variance alpha beta etaLow t q)
```

Lean proof endpoint (built separately)

```
MBJG25ProducerFairness.paper_theorem3_1_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: The target expresses weakly increasing squared bias and weakly decreasing variance as prior strength rises, with strict variance decrease restricted to 0<q<1 and etaLow<etaHigh, exactly matching the approved endpoint correction; positive discrete t and q∈[0,1] are stated approved model context.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 335-344

Verbatim source input

Theorem 3.2. Suppose we have a prior-weighted rating sys- shifted significantly by ratings data. η = 1000 corresponds
tem in the fixed setting with prior parameters (η α̃, η β̃). Con- to a setting where the posterior is essentially equal to the
sider quality estimation after t timesteps. prior throughout the entire simulation, i.e., ratings data is
Then, as a function of true product quality, squared bias ignored. We ran all of our experiments (for both the fixed
[U+0010 control character] [U+0011 control character]2 and responsive models) on an internally-provided
↪ comput-
E[q̂ηα̃,ηβ̃ (v, t)|qv] − qv is convex with a global mini- ing cluster, where the compute power required for our simu-
mum at α̃+α̃
, while variance Var[q̂ηα̃,ηβ̃ (v, t)|qv] is concave lations was negligible.
β̃
with a global maximum at 1/2.

Expanded PaperInterface specification

```
∀ {alpha beta eta : ℝ} {t : ℕ},
  0 < alpha →
    0 < beta →
      0 ≤ eta →
        0 < t →
          (∀ (x y lam : ℝ),
              0 ≤ x →
                x ≤ 1 →
                  0 ≤ y →
                    y ≤ 1 →
                      0 ≤ lam →
                        lam ≤ 1 →
                          MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta eta t
                              (lam * x + (1 - lam) * y) ≤
                            lam * MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta eta t x +
                              (1 - lam) * MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta eta t y) ∧
            (0 ≤ alpha / (alpha + beta) ∧
                alpha / (alpha + beta) ≤ 1 ∧
                  ∀ (q : ℝ),
                    0 ≤ q →
                      q ≤ 1 →
                        MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta eta t
                            (alpha / (alpha + beta)) ≤
                          MBJG25ProducerFairness.paper_conditional_squared_bias alpha beta eta t q) ∧
              (∀ (x y lam : ℝ),
                  0 ≤ x →
                    x ≤ 1 →
                      0 ≤ y →
                        y ≤ 1 →
                          0 ≤ lam →
                            lam ≤ 1 →
                              lam * MBJG25ProducerFairness.paper_conditional_variance alpha beta eta t x +
                                  (1 - lam) * MBJG25ProducerFairness.paper_conditional_variance alpha beta eta t y ≤
                                MBJG25ProducerFairness.paper_conditional_variance alpha beta eta t
                                  (lam * x + (1 - lam) * y)) ∧
                ∀ (q : ℝ),
                  0 ≤ q →
                    q ≤ 1 →
                      MBJG25ProducerFairness.paper_conditional_variance alpha beta eta t q ≤
                        MBJG25ProducerFairness.paper_conditional_variance alpha beta eta t (1 / 2)
```

Lean proof endpoint (built separately)

MBJG25ProducerFairness.paper_theorem3_2_proof

Recorded source-to-Spec screening **Verdict:** matches

Reason: The target formalizes the source claims on true-quality domain [0,1]: squared conditional bias is convex with global minimum at alpha/(alpha+beta), and conditional variance is concave with global maximum at 1/2; positive discrete t is covered by the approved model context.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation